SAI SAMPATH KEDARI

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Work Experience

Abundance Intelligence: Founding Researcher | *San Francisco, CA* **Mar. 2026 – Present**

- Developed a reward-learning framework that converts egocentric human demonstration videos into dense reward signals for training and evaluating robotic manipulation policies.

Peer Robotics: Motion and Controls Intern | *Detroit, MI* **Jul. 2025 – Sep. 2025**

- Developed encoder-based human-force detection for manual takeover and ROS 2 motion-control logic for an autonomous pallet jack towing a passive trailer.

iRaL, University of Michigan: Graduate Researcher | *Ann Arbor, MI* **Aug. 2024 – Jun. 2025**

- Modeled nonlinear residual dynamics in high-speed quadrotor flight using Gaussian processes and learned Koopman operators, enabling the unknown dynamics to be represented in a lifted linear form for model-based control.
- Developed Bayesian state-estimation methods to infer robot states under process and measurement uncertainty for reliable high-speed feedback control.

Dassault Systèmes & Altair Engineering: R&D Software Developer | *India* **Aug. 2019 – Aug. 2021**

- Developed C++ features and APIs for CATIA FTA and HyperMesh, integrating Python and Tcl/Tk.

Open-Source Research Software

Bimanual Robot Learning with OpenArm | *ROS 2, MuJoCo* **In Development**

- Building an end-to-end bimanual manipulation pipeline for a dual-arm 14-DOF OpenArm, integrating MuJoCo simulation, ROS 2 control, and LeRobot-format demonstration collection toward behavior cloning, ACT, and diffusion-policy evaluation on long-horizon tableware tasks.

Deep RL Algorithms Library | GitHub | Derivations | Experiments **In Development**

- Developing a unified PyTorch deep-RL library covering on-policy (REINFORCE, A2C, NPG, TRPO, PPO) and off-policy (DQN, DDPG, TD3, SAC) methods, implementing GAE, Fisher-vector products with conjugate-gradient natural gradients, KL trust-region line search, and clipped surrogate objectives from first principles.
- Benchmarking on MuJoCo locomotion suites (HalfCheetah, Hopper, Walker2d) with multi-seed runs, analyzing sample efficiency, estimator bias–variance, and training stability across algorithm families.

Reinforcement Learning Algorithms Library | GitHub | Derivations | Experiments

- Turned Sutton & Barto into an executable theory-to-code reference, pairing modular NumPy implementations of DP, Monte Carlo, TD, eligibility traces, SARSA, Q-learning, n-step and semi-gradient methods, state aggregation, and tile coding with 17 first-principles derivations extending through policy gradients and GAE.
- Reproduced the book's published benchmark results across Blackjack, Windy Gridworld, Random Walk, and Mountain Car, matching known learning curves in both tabular and continuous-state settings.

Bayesian State & Parameter Estimation | State Estimation | Parameter Inference | Derivations

- Built a Bayesian estimation library for robot state estimation and sensor fusion under nonlinear dynamics: KF, EKF, UKF, Gauss–Hermite, and particle filters from scratch, benchmarked on nonlinear pendulum tracking, with derived optimal proposals, ESS-based resampling, and MCMC-based parameter identifiability analysis.

Monte Carlo Sampling & Inference Library | GitHub | Derivations | Experiments

- Built a Monte Carlo library for computational inference: importance sampling (standard and self-normalized), variance reduction with control variates, and adaptive MCMC (MH, Adaptive Metropolis, Delayed Rejection, DRAM); validated on rare-event estimation and banana-shaped posteriors with IAC and ESS diagnostics.

Education

University of Michigan, Ann Arbor **Aug 2021 – Apr 2024**
M.S. in Mechanical Engineering – Robotics, Mathematics & Statistics
M.S. in Automotive Engineering – Control & Machine Learning

National Institute of Technology Rourkela, India **Jul 2015 – May 2019**
B.Tech. in Mechanical Engineering

Technical Skills

Languages & Tools: C++, Python, PyTorch, NumPy, MuJoCo, Gymnasium, ROS 2, SciPy